

A Bayesian Network for Predicting Diabetic Patient Readmission via Hill-Climb Structure Learning

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Abstract

This project models 30-day hospital readmission risk for diabetic patients as a discrete Bayesian Network whose structure is learned from the UCI *Diabetes 130-US Hospitals* dataset with Hill Climb search. We systematically explore the hyperparameter space of the search itself — structure score, maximum in-degree, number of features, category cardinality — with a 24-configuration grid search selected by validation macro-F1, and find that the scoring function and search depth matter far more than any single feature choice. The winning configuration (K2 score, max in-degree 3, 12 features) clearly outperforms a majority-class baseline on held-out test data, reaching a macro-F1 of 0.312 versus 0.248 for the baseline, while remaining compact and interpretable.

Introduction

Domain

Hospital readmission of diabetic patients is a well-studied problem in clinical informatics: unplanned readmissions are costly and often preventable. We model this domain using the UCI *Diabetes 130-US Hospitals for Years 1999–2008* dataset (Cloutier et al. 2014), which records 101,766 inpatient encounters across 130 US hospitals, labelled with whether the patient was readmitted within 30 days (<30), after 30 days (>30), or not at all (NO). Prior work on this data (Strack et al. 2014) used logistic regression; we instead learn the domain structure automatically with a Bayesian Network.

Aim

The purpose of this project is to learn a Bayesian Network structure for the three-class `readmitted` outcome using Hill Climb search, and — more importantly — to understand *how* the search's own hyperparameters shape the result. Hill Climb is a greedy, score-based procedure, so its outcome depends heavily on choices such as the scoring function and the maximum number of parents allowed per node; rather than picking these by hand, we run a grid search over them and select by validation performance, treating the search configuration itself as the main object of study.

Method

We used the `pgmpy` library (Ankan and Panda 2015) throughout. Before any learning, the raw extract was cleaned following (Strack et al. 2014)'s preprocessing (deduplicating patients, excluding hospice/death discharges, grouping diagnosis codes into 9 clinical categories, mapping coded IDs to labels) to obtain a smaller set of clinically meaningful, low-cardinality columns; this is a one-off, data-independent preparation step and not the focus of this report. We then: (i) split the data 64%/16%/20% into train/validation/test, stratified on the target; (ii) ranked candidate features by mutual information with the target, greedily filtering redundant ones (pairwise normalized mutual information ≥ 0.90); (iii) discretized numeric columns into quantile bins and capped categorical cardinality, mapping rare/unseen values to `Other`; and (iv) fitted CPDs with Bayesian estimation using a BDeu prior (equivalent sample size 10).

The structure itself is learned by **Hill Climb search**: starting from an empty graph, it greedily applies the single edge addition, removal, or reversal that most increases a decomposable structure score, subject to acyclicity and a maximum-in-degree bound that caps how many parents (and hence how large the CPT) a node may have, until no local move improves the score further. We compared the BIC and K2 scores. Because the outcome of this search is sensitive to the scoring function, the in-degree bound, and the feature set it searches over, we ran an exhaustive **grid search** over 24 configurations — number of features $\in \{8, 12, 16\}$, score $\in \{\text{BIC}, \text{K2}\}$, max in-degree $\in \{2, 3\}$, category cap $\in \{8, 12\}$, at a fixed 4 numeric bins — training a candidate network for each and selecting by validation macro-F1 (log-loss as tie-breaker).

Results

The grid search shows the search configuration matters more than any individual feature: every BIC-scored candidate converged to the same macro-F1 of 0.330 regardless of the in-degree bound or category cap, whereas K2 improved from 0.339 (max in-degree 2) to 0.351 (max in-degree 3), and adding features beyond 12 gave no further gain. The winning configuration (12 features, K2, max in-degree 3, category cap 8) was retrained on train+validation and reaches a held-out test macro-F1 of 0.312 (accuracy 0.604), versus 0.248 (accuracy 0.593) for a majority-class baseline.

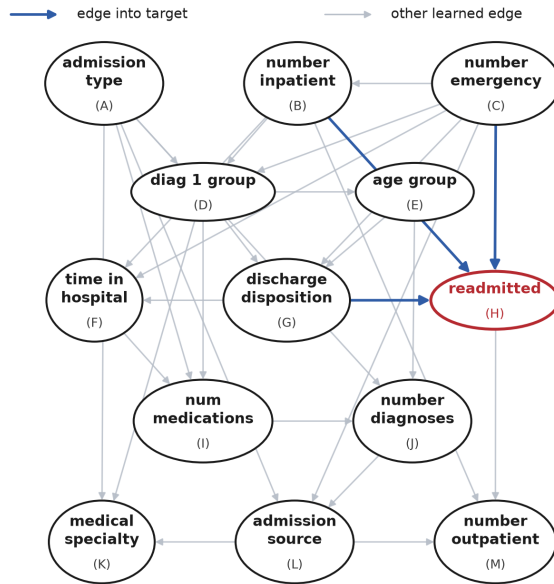


Figure 1: Structure learned by Hill Climb search (K2 score, max in-degree 3) over the 12 selected features, drawn as a hierarchical DAG with sources at the top and sinks at the bottom; edges into the target are highlighted in blue. readmitted is the red target node.

Model

The network (Figure 1) has 12 feature nodes plus the target readmitted, connected by 29 directed edges. The nodes are: number_inpatient, number_outpatient and number_emergency (utilization counts in the prior year), discharge_disposition, admission_type and admission_source (categorical encounter metadata), time_in_hospital, num_medications and number_diagnoses (numeric, quantile-binned into 4 states), medical_specialty and age_group (categorical, capped to the 8 most frequent values plus Other/Missing), and diag_1_group, the primary diagnosis grouped into 9 clinical categories. readmitted itself has three states (<30, >30, NO) and, as learned by Hill Climb, has three direct parents — number_inpatient, number_emergency and discharge_disposition — meaning the other 9 selected features influence the target only indirectly, through the surrounding graph.

No part of the structure or CPDs was manually specified: the DAG in Figure 1 is exactly what Hill Climb search returned for the winning grid-search configuration, and the CPDs are BDeu-smoothed estimates fitted afterwards.

Analysis

Experimental setup

We evaluated the final network, retrained on the combined training and validation splits, on the untouched 20% test split (13,998 encounters), against a majority-class

DummyClassifier baseline trained on the same data. We report accuracy, balanced accuracy, macro/weighted F1, log-loss, macro ROC-AUC/PR-AUC, and the multiclass Brier score. Since the three classes are strongly imbalanced (roughly 59%/32%/9% for NO/>30/<30), we treat macro-F1, not accuracy, as the primary metric, as a trivial majority-class predictor already reaches 59% accuracy. Given the grid-search results, we expected the K2-scored, deeper-search network to beat the baseline by a moderate margin, struggling most on the minority <30 class — the hardest and clinically most actionable class to predict.

Results

On the test split, the network reaches accuracy 0.604, balanced accuracy 0.358 and macro-F1 0.312, versus 0.593, 0.333 and 0.248 for the majority baseline — a clear, if modest, improvement, consistent with the validation-time gain the grid search predicted for this configuration over shallower or BIC-scored alternatives. Per-class F_1 is 0.75 for NO, 0.16 for >30 and only 0.02 for <30 (recall 0.01): the model recognizes non-readmitted patients far better than the rare <30 class, which the confusion matrix confirms is almost always predicted as NO — unsurprising given how small and diffuse that class is (9% of test encounters) relative to the max-in-degree cap and discretization granularity the search was constrained to. As a secondary sanity check, the cleaning stage’s output (69,987 encounters) closely matches the 69,984 reported by (Strack et al. 2014), giving confidence that any remaining error comes from the modelling choices analyzed above rather than a faulty pipeline.

Conclusion

Treating Hill Climb’s own hyperparameters as the primary object of study, rather than fixing them by hand, was informative: the scoring function dominated the outcome (K2 consistently beat BIC, which collapsed to the same structure regardless of the in-degree bound), a deeper in-degree bound further helped K2, and feature count saturated quickly. The resulting network is compact and interpretable and still clearly beats a majority-class baseline, though the gain is concentrated in the majority class and the model struggles with early (<30) readmissions. A natural extension would be to grid-search a class-weighted or two-stage objective directly, rather than macro-F1 alone, to see whether the minority class benefits from the same kind of systematic search.

Links to external resources

References

- Ankan, A., and Panda, A. 2015. pgmpy: Probabilistic graphical models using python.
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